

Program – 4: Integrate Git with Docker for Source-Controlled Application Builds

Description

This experiment demonstrates how to create a Node.js application, containerize it using Docker, push the source code to GitHub, and enable source-controlled application builds.

Prerequisites

- Git installed
- Docker installed
- Node.js and npm installed
- GitHub account

Project Structure

Program-4/

- Dockerfile
- index.js
- package.json
- package-lock.json

Procedure

Step 1: Create Project Folder

```
mkdir program-4  
cd program-4
```

Step 2: Initialize Node.js Project

```
npm init -y
```

Step 3: Create index.js

```
const express = require("express");  
const app = express();  
const PORT = 3000;  
app.get("/", (req, res) => {  
  res.send("Hello from Docker + Git!");  
});  
app.listen(PORT, () => {  
  console.log(`Server running at http://localhost:${PORT}`);  
});
```

Step 4: Install Dependency

```
npm install express
```

Step 5: Create Dockerfile

```
FROM node:18-alpine  
WORKDIR /app  
COPY package*.json ./  
RUN npm install  
COPY . .  
EXPOSE 3000  
CMD ["node", "index.js"]
```



Step 6: Create GitHub Repository

Create a new repository on GitHub (no README).

Step 7: Initialize Git Locally

```
git init
```

```
git add .
```

```
git commit -m "Initial commit"
```

```
git branch -M main
```

Step 8: Link Remote Repository

```
git remote add origin https://github.com/<username>/<repository>.git
```

```
git push -u origin main
```

